

SEMICONDUCTOR TRADE ANALYSIS

U.S. Semiconductor Exports — Full EDA & Findings | HS Code 3542 | 2025

Overview

This document presents an Exploratory Data Analysis of U.S. semiconductor exports under HS Code 3542 (electronic integrated circuits — processors and controllers). The dataset covers 950 export transactions across January through October 2025, spanning 142 importing countries. The analysis seeks to identify which countries receive high-value shipments, whether export value patterns are statistically tied to specific destinations, and how month-to-month volatility varies across major trading partners.

Dataset Summary

Attribute	Value
Total transactions	950
Time period	January – October 2025
Importing countries	142
HS Code	3542 — Electronic integrated circuits (processors and controllers)
Value metric	FOB (Free On Board) in USD
Data source	UN Comtrade

Step 1 — Initial Data Exploration

Objective

Understand the structure and integrity of the dataset before performing any analysis.

Code

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from scipy.stats import chi2_contingency

# Load the cleaned export data
# Latin1 encoding handles special characters in country names (e.g. Turkiye, Curacao)
df = pd.read_csv('Cleaned_HS_3542_Un_Comtrade_USA_Exports.csv', encoding='Latin1')

# Initial exploration
print(df.head())
print(df.shape, df.describe(include='all'), df.info())
```

Key Design Choice

- Latin1 encoding:** Used because some country names contain special characters. This prevents read errors without requiring preprocessing of the source file.

Step 2 — Data Type Validation

Objective

Ensure that numeric columns are correctly typed before any aggregation or statistical analysis.

Code

```
# Convert key columns to numeric (handling any non-numeric values)
for col in ['fobvalue', 'refYear', 'reporterCode']:
    df[col] = pd.to_numeric(df[col], errors='coerce')
```

Key Design Choice

- **errors='coerce'**: Instead of crashing when a non-numeric value is encountered, Python converts it to NaN and continues — allowing problems to be spotted without breaking the analysis.

Step 3 — Distribution Analysis

Objective

Visualize how export values are distributed across all transactions to detect skewness and outliers.

Code

```
# Histogram of FOB export values
ax1 = sns.histplot(df['fobvalue'], bins=30)

# Add count labels on top of each bar
for p in ax1.patches:
    height = p.get_height()
    if height > 0:
        ax1.text(p.get_x() + p.get_width() / 2, height, int(height),
                 ha='center', va='bottom', fontsize='8', rotation=0)

plt.grid(axis='y', linestyle='--')
plt.ticklabel_format(style='plain', axis='x')
plt.show()
```

Key Insight

The distribution is heavily right-skewed. Most exports are in the lower value ranges, but a handful of countries — the major tech manufacturing hubs — account for the truly high-value shipments.

Step 4 — Chi-Square Statistical Test

Objective

Test whether there is a statistically significant relationship between which country imports and what export value tier they receive.

Code

```
# Split FOB values into 3 equal-sized groups
df['fob_cat'] = pd.qcut(df['fobvalue'], q=3, labels=['Low', 'Medium', 'High'])

# Build contingency table: rows = countries, columns = FOB categories
contingency = pd.crosstab(df['partnerISO'], df['fob_cat'])

# Perform chi-square test
chi2, p, dof, expected = chi2_contingency(contingency)
print('Chi-Square Statistic: ', chi2)
```

```
print('P-value: ', p)
print('Degrees of Freedom: ', dof)
```

Statistical Results

Metric	Value
Chi-Square Statistic	1,431.93
P-value	5.14e-153 (essentially zero)
Degrees of Freedom	282

Key Design Choices

- **pd.qcut() vs pd.cut():** qcut() creates equal-sized groups rather than equal-width bins. This prevents empty cells that would invalidate the chi-square test and ensures each category has roughly the same number of observations.

The p-value of 5.14e-153 is statistically definitive: different countries consistently receive different value tiers of semiconductor exports. This validates using country as the primary segmentation variable for trade strategy.

Step 5 — Heatmap Visualization

Objective

Visually confirm which countries cluster in which FOB value categories.

Code

```
Countries = contingency.index.tolist()

# Loop through countries in groups of 10 for readability
for i in range(0, len(Countries), 10):
    group = Countries[i:i+10]
    subset = contingency.loc[group]
    plt.figure(figsize=(12, 6))
    sns.heatmap(subset, annot=True, fmt='d', cmap='Greens')
    plt.title('Contingency Table: Importer Country Vs FOB Category')
    plt.xlabel('FOB Category')
    plt.ylabel('Importer Country')
    plt.show()
```

Key Design Choice

- **Groups of 10:** Processing heatmaps in batches of 10 countries keeps charts readable. Displaying all 142 countries at once would create an incomprehensible visualization.

Key Insight

Major economies (China, Germany, Japan, South Korea) show heavy concentration in the 'High' category. Smaller economies appear predominantly in the 'Low' category — consistent with the demand profiles of large versus small tech manufacturing hubs.

Step 6 — Boxplot Analysis

Objective

Understand the distribution of FOB values within each country, including median, spread, and outliers.

Code

```
with open('boxplt_stats.txt', 'w') as f:
    for i in range(0, len(Countries_list), 10):
        group = Countries_list[i:i+10]
        subset = df[df['partnerISO'].isin(group)]
        if subset.empty:
            continue
        plt.figure(figsize=(12, 6))
        sns.boxplot(x='partnerISO', y='fobvalue', data=subset)
        # Manual quartile calculation
        for country in group:
            values = subset.loc[subset['partnerISO'] == country, 'fobvalue']
            q1 = np.percentile(values, 25)
            q3 = np.percentile(values, 75)
            iqr = q3 - q1
            outliers = values[(values < q1 - 1.5*iqr) | (values > q3 + 1.5*iqr)]
```

Key Findings from Boxplot Statistics

Country	Observation
China	Extremely high median (~\$838M) with massive outliers — consistent high-value exports and occasional mega-transactions
Canada	High median (~\$51M) with moderate variability
Mexico	Moderate median (~\$17M) with some high outliers
Small economies	Very low medians (<\$10K) with minimal variability
Singapore, South Korea	Outliers common, suggesting periodic large orders or special projects

Key Design Choice

- **Manual quartile calculation:** Quartiles, whiskers, and outliers were calculated manually using NumPy rather than relying solely on seaborn's internal calculations. This ensures precise numeric values for documentation and perfect consistency between stats output and visual boxplots.
- **1.5 x IQR rule:** Industry-standard criterion for flagging genuinely unusual values without over-sensitivity to normal variation — the same rule used by default in most statistical software.

Step 7 — Time Series Analysis

Objective

Track how exports change month by month for each country, identifying growth trends, seasonal patterns, and volatility.

Code

```
with open('lineplot_stats.txt', 'w') as f:
    for i in range(0, len(Countries_list), 10):
        group = Countries_list[i:i+10]
```

```
subset = df[df['partnerISO'].isin(group)]
plt.figure(figsize=(12, 6))
ax = sns.lineplot(x='refMonth', y='fobvalue', hue='partnerISO',
data=subset)
# Annotate first, max, and last values on each line
for line in ax.lines:
    x_data = line.get_xdata()
    y_data = line.get_ydata()
    if len(x_data) > 0:
        ax.text(x_data[0], y_data[0], f'{int(y_data[0]):,}',
                ha='center', va='bottom', fontsize=6, rotation=45)
        max_idx = y_data.argmax()
        ax.text(x_data[max_idx], y_data[max_idx],
f'{int(y_data[max_idx]):,}',
                ha='center', va='bottom', fontsize=6, rotation=45)
```

Key Findings from Time Series

- China dominated throughout all 10 months, with values ranging from \$400M to \$900M per month.
- High volatility: Several countries showed significant month-to-month variation, suggesting project-based or cyclical ordering patterns.
- Growth trends: Brazil and India showed upward trajectories, indicating growing semiconductor demand.
- Limited seasonality: With only 10 months of data, definitive seasonal patterns could not be established — some countries showed Q2–Q3 peaks.

Key Design Choice

- **Annotation strategy:** First, maximum, and last values are labeled on each trend line. This allows viewers to quickly identify starting points, peaks, and current status without consulting raw data or squinting at axes.

Summary of Findings

Finding #1 — Extreme Concentration of Export Value

China absolutely dominates U.S. semiconductor exports. The top 10 countries likely represent 80%+ of total export value — a classic Pareto distribution. Most countries (100+) receive relatively small, consistent shipments.

Finding #2 — Statistically Significant Country Patterns

The chi-square test ($p < 0.001$) proved that export value categories are NOT randomly distributed across countries. Different countries have fundamentally different semiconductor demand profiles — validating country as the primary segmentation variable for trade strategy.

Finding #3 — High Variability in Major Markets

Large economies like China, Japan, South Korea, and Germany show high medians but also massive outliers — a mix of routine orders and occasional mega-projects. Smaller economies show much more consistent but lower order patterns.

Finding #4 — Limited Temporal Patterns

With only 10 months of data, strong seasonality trends could not be definitively established. Month-to-month volatility was evident, particularly in major markets. Year-over-year analysis would require additional data from previous years.

Analysis Limitations

- Only 10 months of data limits trend analysis and seasonality detection.
- FOB value does not account for quantity, making per-unit pricing analysis impossible.

- No data on rejected orders, delayed shipments, or trade barriers.

Files Generated During This Analysis

File	Description
boxplt_stats.txt	Comprehensive quartile, whisker, and outlier statistics for all 142 countries
lineplot_stats.txt	First, maximum, last, and month-by-month values for temporal analysis
Visualizations	Histograms, heatmaps, boxplots, and line plots saved during execution

Document Metadata

- **Data Source:** [UN Comtrade](#)
- **HS Code:** 3542 — Electronic integrated circuits (processors and controllers)
- **Analysis Tools:** Python (pandas, numpy, seaborn, matplotlib, scipy)
- **Output Files:** boxplt_stats.txt, lineplot_stats.txt, visualizations